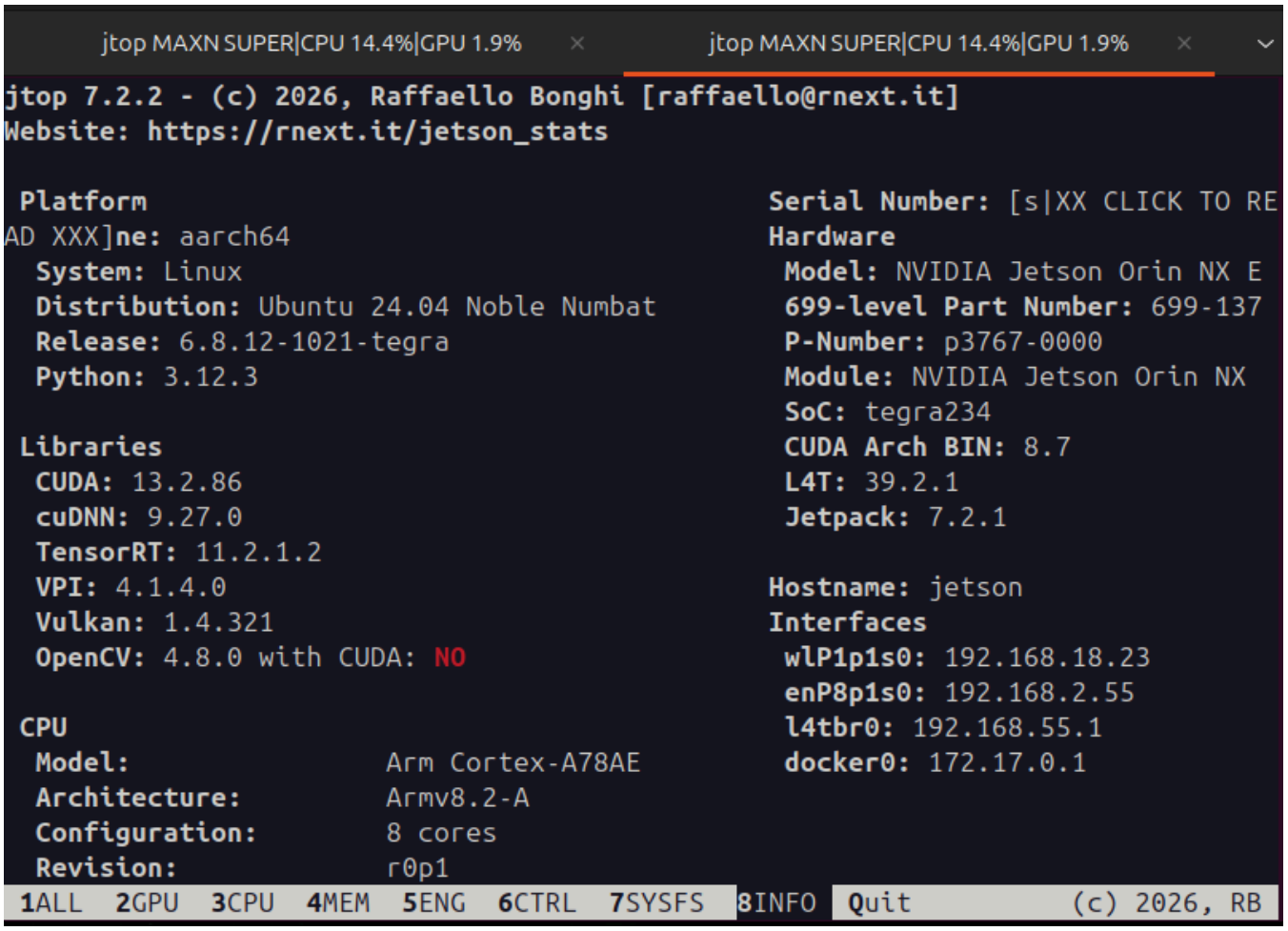
YOLO environment construction

YOLO environment construction

1. The system information
2. Download the files
3. Install Ultralytics
4. Configure the GPU acceleration
5. Verify the installation

References

1. The system information



2. Download the files

Note: if the environment has already been built following YOLO11, the ultralytics folder of the home directory has to be renamed first, and then the files are downloaded.

Here, downloading ultralytics-v8.4.21 is taken as an example

```
cd ~/
git clone https://github.com/ultralytics/ultralytics.git -b v8.4.21 --depth=1
```

```
jetson@ubuntu:~$ git clone https://github.com/ultralytics/ultralytics.git -b v8.
4.21 --depth=1
Cloning into 'ultralytics'...
remote: Enumerating objects: 979, done.
remote: Counting objects: 100% (979/979), done.
remote: Compressing objects: 100% (827/827), done.
remote: Total 979 (delta 209), reused 290 (delta 137), pack-reused 0 (from 0)
Receiving objects: 100% (979/979), 2.44 MiB | 59.00 KiB/s, done.
Resolving deltas: 100% (209/209), done.
Note: switching to '83471a5041fe5d7fdba5c12aff65f4f84f27c081'.

You are in 'detached HEAD' state. You can look around, make experimental
changes and commit them, and you can discard any commits you make in this
state without impacting any branches by switching back to a branch.
```

If the repository cannot be cloned due to network problems, you can go to the following URL to download the archive file, and then extract it.

```
https://github.com/ultralytics/ultralytics/releases/tag/v8.4.21
```

3. Install Ultralytics

```
cd ~/ultralytics
sudo pip3 install ".[export]" -i https://pypi.tuna.tsinghua.edu.cn/simple
```

```
jetson@ubuntu:~/ultralytics$ sudo pip3 install ".[export]" -i https://pypi.tuna.
tsinghua.edu.cn/simple
Looking in indexes: https://pypi.tuna.tsinghua.edu.cn/simple
Processing /home/jetson/ultralytics
  Installing build dependencies ... done
  Getting requirements to build wheel ... done
  Preparing metadata (pyproject.toml) ... done
Requirement already satisfied: numpy>=1.23.0 in /usr/local/lib/python3.10/dist-p
ackages (from ultralytics==8.4.21) (1.26.4)
Requirement already satisfied: matplotlib>=3.3.0 in /usr/local/lib/python3.10/di
st-packages (from ultralytics==8.4.21) (3.10.8)
Requirement already satisfied: opencv-python>=4.6.0 in /usr/local/lib/python3.10
/dist-packages (from ultralytics==8.4.21) (4.11.0.86)
Requirement already satisfied: pillow>=7.1.2 in /usr/lib/python3/dist-packages (
from ultralytics==8.4.21) (9.0.1)
```

4. Configure the GPU acceleration

Since torch, torchvision and onnxruntime-gpu have already been installed in our previous tutorials, there is no need to run the installation commands here.

> **torch**

> **torchvision**

> **onnxruntime-gpu**

5. Verify the installation

> Verify Ultralytics

```
python3 -c "import ultralytics; print(ultralytics.__version__)"
```

> Verify Torch

```
python3 -c "import torch; print(torch.__version__); print(torch.cuda.is_available())"
```

> Verify Torchvision

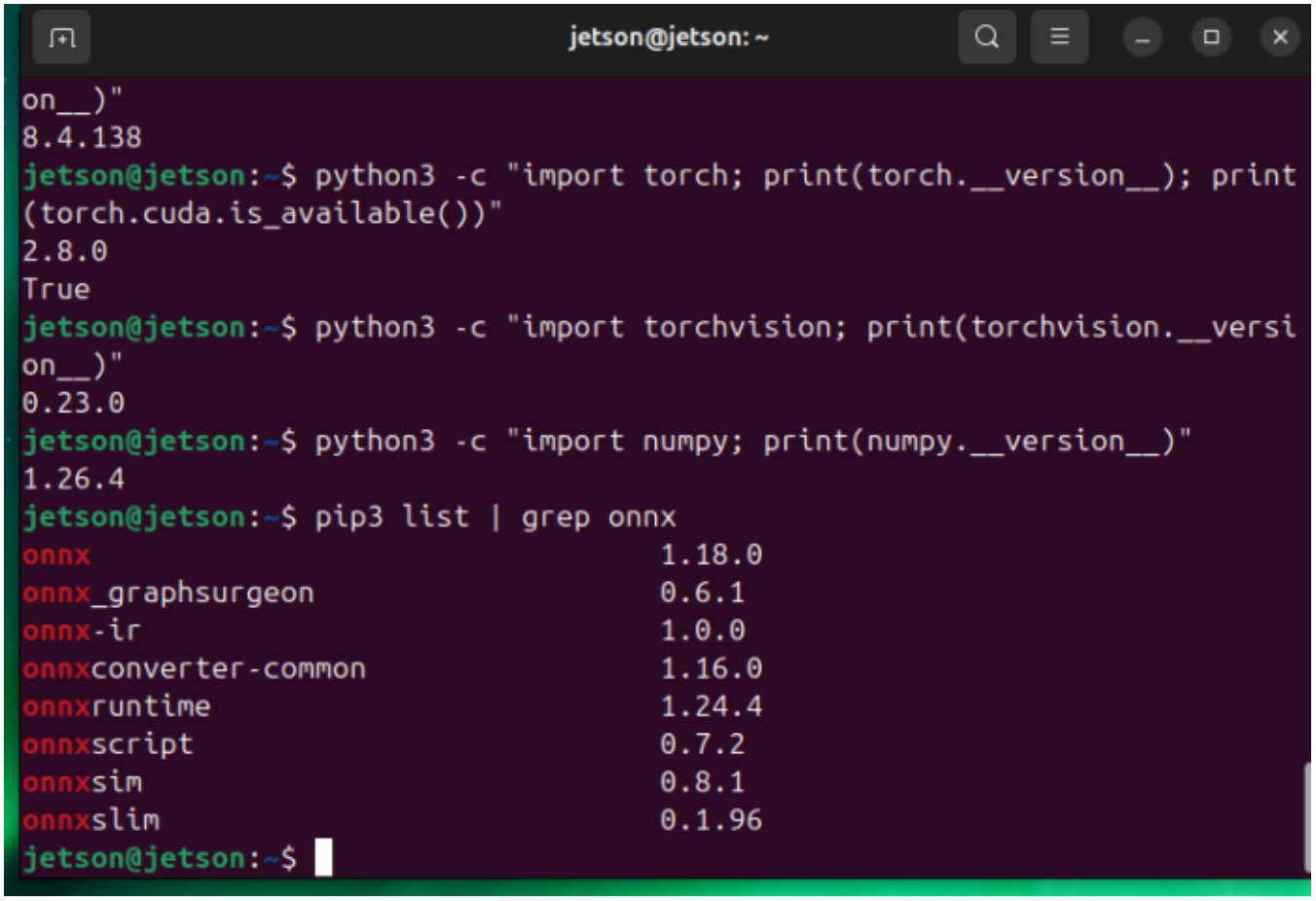
```
python3 -c "import torchvision; print(torchvision.__version__)"
```

> Verify Numpy

```
python3 -c "import numpy; print(numpy.__version__)"
```

> Verify onnxruntime-gpu

```
pip3 list | grep onnx
```



References

https://docs.ultralytics.com/guides/nvidia-jetson/